

Does the Weather Still Move Markets?

A Pre-Registered, Two-Hemisphere, Post-Publication Test

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Abstract

We re-examine weather and seasonal-mood effects in stock returns with two pre-registered studies, free and replicable data, and a placebo-city inference procedure that directly addresses the concern that weather effects are data-mined. Study 1 uses daily US (1955–2026) and Australian (1984–2026) returns. After replicating [Hirshleifer and Shumway \(2003\)](#) and [Kamstra et al. \(2003\)](#) on their own samples, one of nine primary tests survives multiple-testing correction and placebo inference: New York cloud cover lowers same-day US returns by about 2 basis points per standard deviation (Holm $p = 0.029$); none of 100 placebo cities produces an effect as large. Study 2, pre-registered after Study 1, pools 26 further exchanges and seven weather variables. Rain days lower returns (1.6 basis points; Holm $p = 0.045$, beyond all placebo draws), and high air pressure and sunshine point the same way. However, the international cloud effect is absent after the [Hirshleifer and Shumway \(2003\)](#) sample ends, and every fair-weather effect is concentrated before 2003. Seasonal affective disorder effects do not appear in the southern hemisphere, and temperature does not move markets. Weather effects were small, and evidence that they persisted after publication is weak; a simple cloud-timed trading strategy is not profitable after plausible transaction costs.

Keywords: weather, investor mood, seasonal affective disorder, pre-registration, replication, placebo inference

JEL: G12, G14, G41

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1 Introduction

Few findings in behavioural finance are as memorable, or as contested, as the claim that sunshine lifts stock prices. [Saunders \(1993\)](#) reported that New York Stock Exchange returns were lower on cloudy days in New York; [Hirshleifer and Shumway \(2003\)](#) extended the result to 26 exchanges; [Kamstra et al. \(2003\)](#) argued that seasonal affective disorder (SAD) produces a seasonal cycle in returns that reverses between hemispheres; and [Cao and Wei \(2005\)](#) linked low temperatures to higher returns. Each finding was challenged soon after publication. [Trombley \(1997\)](#) showed that the New York effect was sensitive to the sample and the definition of cloudiness; [Jacobsen and Marquering \(2008\)](#) argued that SAD and temperature effects are hard to distinguish from the Halloween effect of [Bouman and Jacobsen \(2002\)](#); [Kelly and Meschke \(2010\)](#) questioned the construction of the SAD variable; and [Novy-Marx \(2014\)](#) showed that a researcher free to choose predictors can find “significant” relations with sunspots and planetary alignments. For Australia, the evidence has always been weak ([Keef and Roush, 2007](#); [Worthington, 2009](#)).

Two decades later the question can be asked in a way that was not possible when these papers were written. More than twenty years of data have accumulated since publication, which gives a genuine out-of-sample test in the spirit of [McLean and Pontiff \(2016\)](#). Pre-registration makes it possible to commit to specifications before seeing the new data. And free, long, hourly weather records for any city on Earth make it possible to ask how unusual the New York result is relative to the weather in cities that have nothing to do with the US market.

This paper makes four contributions. First, it pre-registers nine primary tests covering cloud cover, temperature and SAD in the US and Australia, their post-publication persistence, a market-structure channel and a small-stock channel, together with a decision rule that requires both a multiple-testing-adjusted p -value and a placebo percentile. Second, it introduces a placebo-city test: the same regression is estimated with weather from 100 cities outside the US and Australia, and the real city’s coefficient is ranked within that distribution. Because the placebo cities share the seasonal structure of weather but not any plausible link to the market, the test speaks directly to the data-mining critique. Third, it uses the reversal of seasons between hemispheres to separate a seasonal-mood channel from calendar effects. Fourth, a second study, pre-registered after the first study’s results were known and run on 26 exchanges whose returns had not been examined, tests seven weather variables (cloud, sunshine, temperature, wind, rain, humidity and pressure) and gives an out-of-sample test of the international results of [Hirshleifer and Shumway \(2003\)](#).

The results are mixed in an informative way. The New York cloud effect survives: -2.1 basis points (bp) per standard deviation of cloud cover over 1955–2026, significant after Holm and Romano–Wolf corrections, and more extreme than every one of 100 placebo cities. But the effect is small, imprecisely estimated after 2003, not visible in Australian data after 2003, and a simple strategy based on it loses money after costs. SAD does not appear in Australia, and temperature does not move the market, although warm days lower the returns of energy and utility stocks, consistent with a physical demand channel rather than mood. Across 26 further exchanges, rain days lower returns and high pressure and sunshine raise them, with rain surviving the full decision rule; but the evidence for these effects comes mainly from before 2003, and the international cloud effect is not detectable after 1997.

2 Hypotheses

Table 2 lists the nine primary tests; the full pre-registration, including the power analysis and decision rule, is in the online repository. All tests are one-sided in the direction the original literature predicts.

- H1** Deseasonalised cloud cover in the exchange city lowers same-day index returns (New York for the US, Sydney for Australia).
- H2** The cloud effect survives publication: its coefficient after July 2003 is negative.
- H3** In Australia, the SAD effect follows the local (southern-hemisphere) season rather than the New York calendar.
- H4** A temperature anomaly lowers returns (Cao and Wei, 2005).
- H5** The New York weather effect weakens after the NYSE Hybrid Market (January 2007) moved most trading off the floor.
- H6** Cloud effects are stronger in small stocks, whose investors are more local and more retail (Loughran and Schultz, 2004).

A primary hypothesis is *supported* only if its one-sided HAC p -value survives a Holm correction at family-wise $\alpha = 0.05$ and the placebo-city percentile lies in the predicted 5% tail; *inconclusive* if only one holds; and *not supported* otherwise. With daily return volatility near 1% and about 5,800 post-2003 days, the minimum detectable effect at 80% power is roughly 3.3 bp per standard deviation, about the size of the original estimates. A post-publication null is therefore informative only through its confidence interval, which we report throughout.

3 Data

Returns. US returns are the daily Center for Research in Security Prices (CRSP) value-weighted market return (market excess return plus the risk-free rate) from Kenneth French’s data library (French, 2026); size and industry portfolios come from the same source. Australian returns are the S&P/ASX All Ordinaries price index from 1984 (Yahoo Finance, 2026). No free daily total-return series covers 1984–2019, and the Reserve Bank of Australia table named in our pre-registration no longer reports dividend yields, so the primary series adds back dividends as the average daily dividend return in each calendar month, estimated from the S&P/ASX 200 total-return and price indices over 2019–2026 (deviation D3, fixed before results were seen).

This approximation cannot matter for the weather tests, for a simple reason. The add-back is a deterministic function of the calendar month, so it can bias a weather coefficient only through a correlation between the weather anomaly and the calendar month. Our anomalies are deseasonalised by week of year, which makes that correlation close to zero by construction: -0.006 for cloud and -0.011 for temperature. Appendix Table 10 confirms this directly. Every Australian weather coefficient is unchanged, to within 0.1 bp, with price-only returns or with calendar-month fixed

effects, which absorb any month-specific add-back exactly. The one test that is seasonal by design, SAD (H3), does depend on the add-back: -1.4 bp with dividends and -0.4 bp without. Neither version is significant, and both have the wrong sign, so the conclusion is the same. All returns are log returns in percent, and the sample ends on 31 August 2026.

Weather. Hourly observations come from NOAA’s Global Historical Climatology Network hourly data set (NOAA National Centers for Environmental Information, 2026) for New York LaGuardia (with JFK filling gaps), and from NOAA’s Integrated Surface Database (Smith et al., 2011) and the Iowa Environmental Mesonet METAR archive (Iowa Environmental Mesonet, 2026) for Sydney Airport. Cloud cover is total sky cover in oktas, and temperature is air temperature; both are averaged over 06:00–16:00 local time, following Hirshleifer and Shumway (2003). Each variable is deseasonalised against a trailing 30-year climatology for the same ISO week, using only earlier years, and standardised. The resulting anomalies are available from 1955 in New York and 1984 in Sydney. For placebo cities, alternative cities and a single-source robustness check we use the ERA5 reanalysis (Hersbach et al., 2020), point time series from the Copernicus Climate Data Store (Copernicus Climate Change Service, 2026), processed identically. Exact definitions are in Appendix D. ERA5 cloud cover correlates 0.85 with the LaGuardia station series at the daily frequency.

Seasonal variables. Night length is computed astronomically for each exchange city’s latitude. SAD_t equals night length minus 12 hours during the local autumn and winter, and zero otherwise; $FALL_t$ is a local-autumn dummy (Kamstra et al., 2003).

4 Method

Each test is an OLS regression of the daily return on the weather or seasonal variable and a fixed set of controls: day-of-week dummies, turn-of-year, pre- and post-holiday, and tax-year-end dummies (December in the US, June in Australia), and two lags of the dependent variable. Australian regressions also control for the most recent completed US return, because the New York session closes before Sydney opens. Standard errors are Newey–West with five lags (Newey and West, 1987). Multiplicity across the nine tests is handled with the Holm procedure (Holm, 1979), with a Romano–Wolf stepdown (Romano and Wolf, 2005) based on a joint stationary bootstrap (Politis and Romano, 1994) as robustness. Bandwidths, bootstrap settings, random seeds, the handling of missing data and holidays, and the exact standardisation formula are given in Appendix D. Every table and figure can be regenerated from the raw downloads with a single command (`make all`).

The placebo-city test re-estimates each weather regression 100 times, each time using ERA5 cloud or temperature anomalies for a different city more than 1,000 km from both New York and Sydney (the list was fixed in the pre-registration). The real city is re-estimated with ERA5 data as well, so the comparison is like for like. We report the number of placebo coefficients at least as extreme. With 100 placebo cities, the resolution of this test is one percentage point, so “0/100” means an empirical placebo percentile below 1%, not a probability of zero. A second placebo shifts

the real station’s weather series by $k = 1, \dots, 30$ years. This preserves seasonality exactly and breaks only the day-to-day link.

5 Results

5.1 Replication

Table 1 shows that our data reproduce the original findings on the original samples. In a logit of up days on deseasonalised New York cloud cover over 1982–1997, the coefficient is -0.114 ($t = -3.56$), matching the sign and significance of [Hirshleifer and Shumway \(2003\)](#). Over 1928–2000, the SAD coefficient is positive (2.2 bp per hour of night beyond 12, $t = 2.77$) and the autumn dummy negative (-5.0 bp, $t = -2.31$), as in [Kamstra et al. \(2003\)](#). Any failure of the extended tests therefore cannot be blamed on data that cannot reproduce the originals.

Table 1: Replication gate (data up to 2000). Pass = predicted sign with one-sided $p < 0.10$.

Specification	Pred.	β (bp)	t	p	N
HS 2003 NYC logit (1982-97)	<0	-11.41	-3.56	<0.001	4,012
HS 2003 NYC OLS (1982-97)	<0	-2.01	-1.26	0.105	4,012
KKL 2003 US SAD (1928-2000)	>0	+2.15	+2.77	0.003	19,411
KKL 2003 US FALL (1928-2000)	<0	-5.01	-2.31	0.011	19,411
KKL 2003 US SAD + weather (1955-2000)	>0	+2.63	+3.00	0.001	9,389
KKL 2003 US FALL + weather (1955-2000)	<0	-3.34	-1.46	0.072	9,389

5.2 Primary tests

Table 2 reports the nine pre-registered tests. Only H1 in the US is supported. A one-standard-deviation increase in New York cloud cover lowers the same-day market return by 2.1 bp (95% CI $[-3.7, -0.6]$). The Holm-adjusted p -value is 0.029 and the Romano–Wolf p -value 0.035. Figure 1 shows the placebo distribution: centred almost exactly on zero (mean -0.03 bp, standard deviation 0.66 bp), with the New York ERA5 coefficient (-1.9 bp) beyond every placebo city. None of the 30 year-shifted series produces a coefficient as negative either.

In Australia the Sydney cloud coefficient has the same sign and similar size (-1.8 bp), and only 2 of 100 placebo cities are as extreme, but it does not survive the Holm correction, so H1-AU is inconclusive.

After publication. The post-2003 US coefficient is -2.0 bp, almost identical to the full-sample estimate, but its confidence interval ($[-5.2, +1.3]$) cannot distinguish persistence from disappearance. For Australia the post-2003 coefficient is $+0.1$ bp with a much tighter interval ($[-2.3, +2.4]$). Figure 2 shows trailing ten-year estimates. The New York coefficient is negative in most windows but close to zero for windows ending 2005–2012. The Sydney coefficient drifts from about -3 bp in the 1990s towards zero after 2015.

Table 2: Pre-registered primary tests. β in basis points per one-standard-deviation weather shock (for H3, per hour of night beyond 12). One-sided HAC p -values in the predicted direction; Holm and Romano–Wolf adjust across the nine tests. Placebo = number of the 100 placebo cities with a coefficient at least as extreme (ERA5); Shift = number of the 30 year-shifted weather series at least as extreme.

ID	Hypothesis	β	95% CI	t	p_1	Holm	R-W	Placebo	Shift	Verdict
H1-US	NYC cloud lowers US returns	-2.13	[-3.7, -0.6]	-2.72	0.003	0.029	0.035	0/100	0/30	Supported
H1-AU	Sydney cloud lowers AU returns	-1.77	[-3.6, +0.1]	-1.87	0.031	0.246	0.224	2/100	1/30	Inconclusive
H2-US	US cloud effect after 2003	-1.95	[-5.2, +1.3]	-1.18	0.119	0.834	0.619	17/100	5/30	Not Supported
H2-AU	AU cloud effect after 2003	+0.07	[-2.3, +2.4]	+0.06	0.523	1.000	0.982	50/100	16/30	Not Supported
H3-AU	AU SAD follows the season	-1.38	[-3.9, +1.2]	-1.06	0.855	1.000	0.982	–	–	Not Supported
H4-US	NYC temperature lowers returns	+0.13	[-1.2, +1.5]	+0.19	0.574	1.000	0.982	50/100	18/30	Not Supported
H4-AU	Sydney temperature lowers returns	+0.67	[-0.9, +2.2]	+0.85	0.803	1.000	0.982	87/100	28/30	Not Supported
H5-US	Effect weakens after NYSE Hybrid	+0.18	[-3.9, +4.3]	+0.09	0.465	1.000	0.982	36/100	18/30	Not Supported
H6-US	Stronger in small minus big	-0.07	[-1.2, +1.1]	-0.11	0.454	1.000	0.982	10/100	12/30	Not Supported

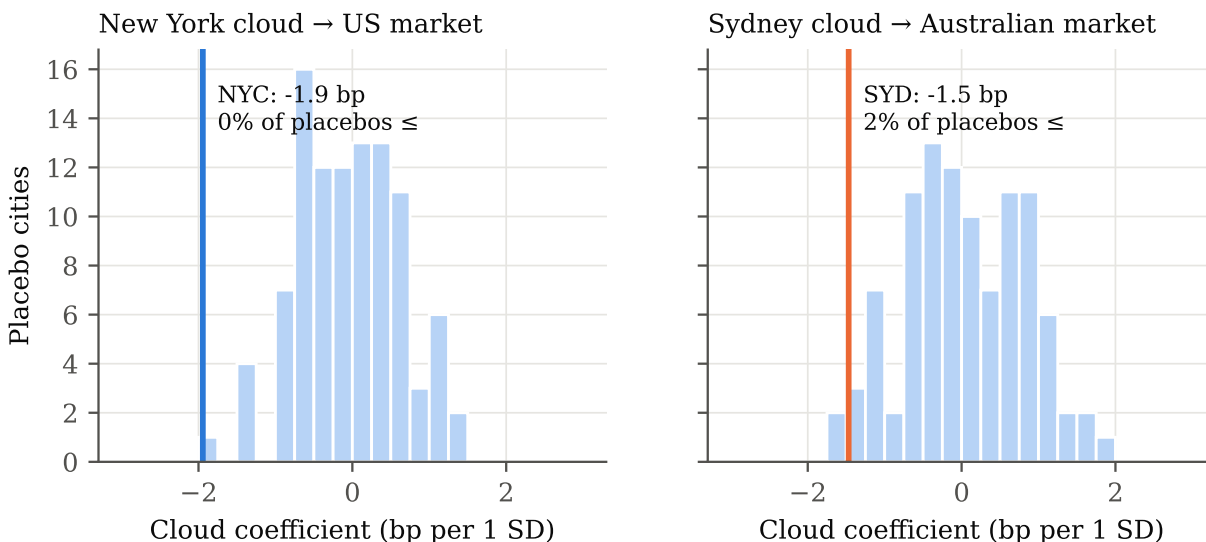


Figure 1: Placebo-city distributions for the cloud coefficient (H1). Bars show coefficients estimated with ERA5 cloud anomalies from 100 placebo cities; the vertical line is the exchange city, also measured with ERA5.

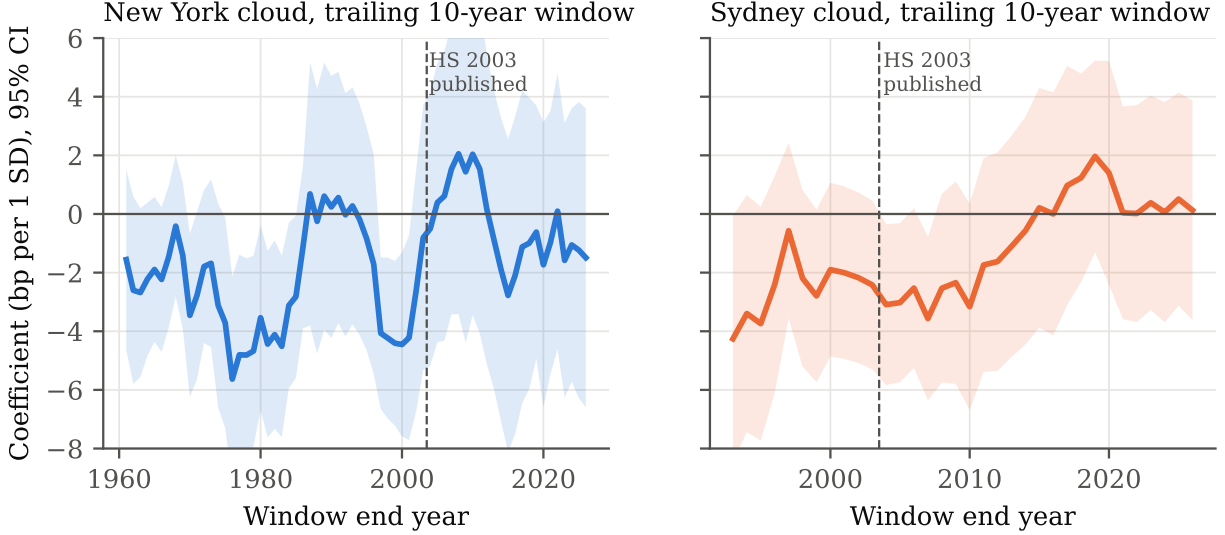


Figure 2: Trailing ten-year estimates of the cloud coefficient, with 95% HAC confidence bands. The dashed line marks the publication of [Hirshleifer and Shumway \(2003\)](#).

Seasons, temperature and channels. The Australian SAD coefficient has the wrong sign (-1.4 bp per hour, $t = -1.06$), so the seasonal-mood cycle does not follow the southern-hemisphere season (H3). Temperature anomalies have no detectable effect in either market (H4). There is no evidence that the New York effect weakened when floor trading declined (H5), although the interaction is imprecise, nor that it is concentrated in small stocks (H6).

5.3 Robustness and exploratory results

Table 6 (appendix) varies the US and Australian cloud and temperature specifications. The US cloud result holds with ERA5 weather (-1.9 bp), full-sample deseasonalisation (-2.1 bp), a logit of up days ($t = -3.73$) and a GARCH(1,1) model with Student- t errors (-2.1 bp, $t = -4.24$). It is weaker with Chicago weather (-1.2 bp, $t = -1.61$), as expected if the relevant weather is where the market is. In a split sample, the coefficient is -2.1 bp ($t = -2.53$) over 1955–2006 and -1.9 bp ($t = -0.97$) over 2007–2026. Rain during trading hours lowers US returns by 5.4 bp ($t = -2.86$), consistent with the cloud result.

The Australian cloud coefficient is robust to price-only returns and ERA5 data, but Melbourne weather shows nothing ($+0.2$ bp). Every post-2003 Australian variant is within 0.6 bp of zero.

Table 7 examines seasonal effects. In the US, the SAD coefficient is positive over 1955–2026 (2.0 bp, $t = 2.40$) and similar but insignificant after 2003. In Australia, neither the local SAD variable (0.2 bp) nor the New York-calendar SAD variable (-0.2 bp) nor a Halloween dummy (1.4 bp, $t = 0.86$) is significant, so the reversed-season test favours neither the mood nor the calendar account. It simply finds no seasonal cycle.

Table 9 provides suggestive evidence for a physical channel. Controlling for the same-day market return, warmer-than-normal days lower the returns of the oil industry (-2.0 bp, $t = -2.87$) and of

utilities (-1.0 bp, $t = -2.25$). This is consistent with lower heating demand rather than investor mood. Cloudy days raise coal-industry returns (5.6 bp, $t = 3.82$). These are sixteen exploratory tests, so only the coal and oil results clear a Bonferroni threshold.

5.4 Economic significance

Table 3 asks whether the effect can be traded. Using the SPY exchange-traded fund from 1993, we hold the market from open to close only on days whose early-morning (06:00–10:00) New York cloud cover is below normal. The open-to-close return on sunny mornings exceeds that on cloudy mornings by only 0.3 bp per day ($t = 0.13$). With a 1 bp round-trip cost, every variant loses money. This specified cloud-timed SPY strategy is therefore not profitable after plausible round-trip transaction costs. We have not searched over other implementations (futures, other markets, other weather signals), and we do not claim that no weather-based strategy could ever be profitable. The result is consistent with the conclusion of [Hirshleifer and Shumway \(2003\)](#) that the effect is too small to trade.

Table 3: Cloud-timed SPY open-to-close strategy, 1993–2026. Long SPY from the open to the close on days whose 06:00–10:00 New York cloud anomaly is below zero (“sunny”), or above it (“cloudy”); otherwise in cash (zero return). Costs are per round trip.

Rule	Cost (bp)	Ann. return (%)	Sharpe	Days held
every day	0	+0.71	+0.05	8,430
sunny mornings only	0	+0.45	+0.05	3,381
cloudy mornings only	0	+0.25	+0.02	5,049
every day	1	-1.81	-0.12	8,430
sunny mornings only	1	-0.56	-0.06	3,381
cloudy mornings only	1	-1.25	-0.11	5,049
every day	2	-4.33	-0.29	8,430
sunny mornings only	2	-1.57	-0.17	3,381
cloudy mornings only	2	-2.76	-0.23	5,049
Sunny minus cloudy mornings: $+0.28$ bp per day ($t = 0.13$), $N = 8,430$				

6 Study 2: Seven weather variables across 26 exchanges

6.1 Design

Study 2 was pre-registered ([prereg/study2_preregistration.md](#)) after Study 1 was complete, and frozen before any return from the new exchanges was examined. We began with a list of 37 national benchmark indices and kept the 26 whose free daily history starts by 2006 and has fewer than 5% zero-change days: twelve in Europe, nine in Asia, four in the Americas and one in New Zealand. Appendix C gives the eligibility algorithm, the full candidate list and the reason for each exclusion. New York and Sydney are excluded from the primary tests because their results were already known. For every city, seven weather variables come from ERA5 and are processed identically: mean cloud cover, solar radiation (“sunshine”), temperature, wind speed, relative humidity and sea-level pressure, each as a trailing-climatology z -score over 06:00–16:00

local time, plus a rain-day indicator (≥ 1 mm) measured relative to its climatological frequency. Returns are scaled by each index’s trailing 60-day volatility, so coefficients are in percent of a daily standard deviation, and estimated in a pooled regression with exchange fixed effects and Driscoll–Kraay standard errors, which allow for correlation across markets on the same day. The decision rule is the same as in Study 1. The placebo test gives each exchange the weather of a random city more than 1,000 km away, 200 times.

6.2 Results

Table 4 reports the nine primary tests. Rain is supported: a rain day lowers returns by 1.6% of a daily standard deviation (about 1.6 bp at the median volatility; Holm $p = 0.045$), and none of the 200 placebo assignments produces an effect as large (empirical percentile below 0.5%). High pressure (+1.0 bp per standard deviation, Holm $p = 0.070$), sunshine (+0.6 bp), cloud (−0.4 bp) and humidity (−0.4 bp) all lie in the extreme tail of their placebo distributions but do not survive the Holm correction, so they are inconclusive. Their signs are consistent: fair weather goes with higher returns. Temperature, wind and SAD show nothing. The SAD test has little power, however, since only four exchanges are in the southern hemisphere.

The out-of-sample cloud test (S2) is clearly negative. Using only data from 1998, after the [Hirshleifer and Shumway \(2003\)](#) sample ends, the pooled cloud coefficient is −0.2 bp ($t = -0.57$) and sits in the middle of its placebo distribution.

Table 4: Study 2 pre-registered primary tests, pooled over 26 exchanges not used in Study 1. β is in percent of a daily return standard deviation per one-SD weather shock (rain: per rain day relative to its climatological frequency); bp converts at the median daily volatility. Driscoll–Kraay standard errors. Placebo = number of the 200 random placebo-city assignments at least as extreme.

ID	Weather variable	β (% SD)	bp	t	p	Holm	Placebo	N	Verdict
S1	Cloud (full sample)	-0.39	-0.41	-1.49	0.068	0.410	1/200	222,941	Inconclusive
S2	Cloud, 1998 onward	-0.16	-0.17	-0.57	0.284	1.000	34/200	184,234	Not Supported
S3	Sunshine	+0.55	+0.58	+2.14	0.016	0.113	2/200	222,941	Inconclusive
S4	Temperature	-0.12	-0.13	-0.39	0.349	1.000	78/200	222,941	Not Supported
S5	Wind	-0.11	-0.12	-0.41	0.342	1.000	110/200	222,941	Not Supported
S6	Rain day	-1.55	-1.64	-2.58	0.005	0.045	0/200	222,941	Supported
S7	Humidity (two-sided)	-0.42	-0.44	-1.66	0.097	0.484	5/200	222,941	Inconclusive
S8	Pressure (two-sided)	+0.93	+0.98	+2.62	0.009	0.070	0/200	222,941	Inconclusive
S9	SAD follows local season	+0.13	+0.14	+0.22	0.412	1.000	–	222,941	Not Supported

Table 5 and Figure 3 show why. The evidence for every fair-weather effect comes mainly from the period before July 2003. The rain coefficient falls from −3.8% of a standard deviation ($t = -3.9$) to −0.3% ($t = -0.4$), sunshine from +1.8% ($t = 4.2$) to −0.1%, and cloud from −1.2% ($t = -2.6$) to zero. Pressure declines least, from +1.5% to +0.7% ($t = 1.5$). The post-2003 intervals are wide enough that smaller persistent effects cannot be excluded; what the data show is that the effects are not detectable after 2003, not that they are exactly zero. The effects are found in developed rather than emerging markets and in the northern hemisphere. When all seven variables enter one regression, only pressure remains significant ($t = 2.1$). This suggests that the fair-weather pattern reflects high-pressure weather systems more than any single variable.

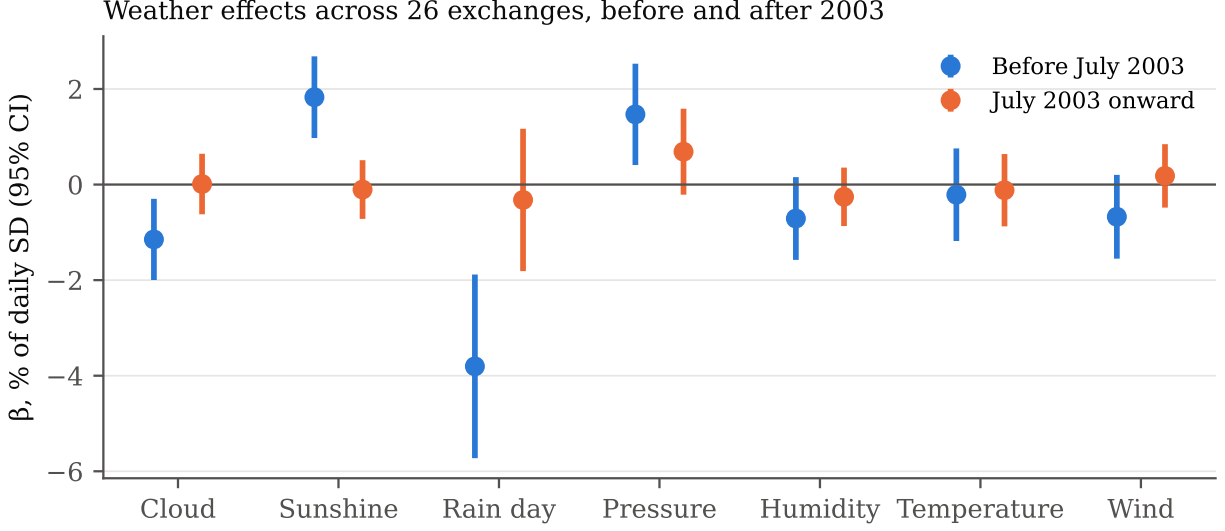


Figure 3: Pooled Study 2 coefficients before and after July 2003, with 95% Driscoll–Kraay confidence intervals.

Table 5: Study 2 exploratory splits (26 exchanges). Cells: β in percent of a daily SD, with t -statistics in parentheses. “Joint” enters all seven anomalies in one regression.

Variable	Pre-2003	Post-2003	Developed	Emerging	Northern	Southern	Joint
Cloud	-1.15 (-2.6)	+0.01 (+0.0)	-0.51 (-1.5)	-0.18 (-0.5)	-0.46 (-1.6)	-0.00 (-0.0)	+0.08 (+0.2)
Sunshine	+1.83 (+4.2)	-0.10 (-0.3)	+0.71 (+2.1)	+0.28 (+0.8)	+0.60 (+2.2)	+0.24 (+0.4)	+0.33 (+0.8)
Temperature	-0.21 (-0.4)	-0.12 (-0.3)	-0.21 (-0.5)	+0.05 (+0.1)	-0.07 (-0.2)	-0.32 (-0.5)	-0.09 (-0.2)
Wind	-0.67 (-1.5)	+0.18 (+0.5)	-0.31 (-0.8)	+0.22 (+0.6)	-0.11 (-0.4)	-0.10 (-0.2)	+0.13 (+0.5)
Rain day	-3.80 (-3.9)	-0.32 (-0.4)	-1.71 (-2.2)	-1.20 (-1.3)	-1.85 (-2.8)	+0.15 (+0.1)	-0.76 (-1.0)
Humidity	-0.71 (-1.6)	-0.26 (-0.8)	-0.53 (-1.5)	-0.24 (-0.7)	-0.46 (-1.6)	-0.26 (-0.4)	+0.02 (+0.0)
Pressure	+1.47 (+2.7)	+0.69 (+1.5)	+1.64 (+3.4)	-0.17 (-0.4)	+1.09 (+2.8)	+0.00 (+0.0)	+0.81 (+2.1)

Figure 4 shows the exchange-level estimates. Twelve of the 196 exchange-variable tests survive a Benjamini–Hochberg false-discovery rate of 10%. Pressure effects appear in London, Frankfurt, Paris, Zurich and Brussels, but these cities share weather systems, so they are closer to a single European finding than to five. Tokyo and Taipei show cloud and sunshine effects like New York’s. Among the other exploratory results, storm and heat days have no effect on returns, and warmer-than-normal days are associated with lower volatility ($t = -2.8$).

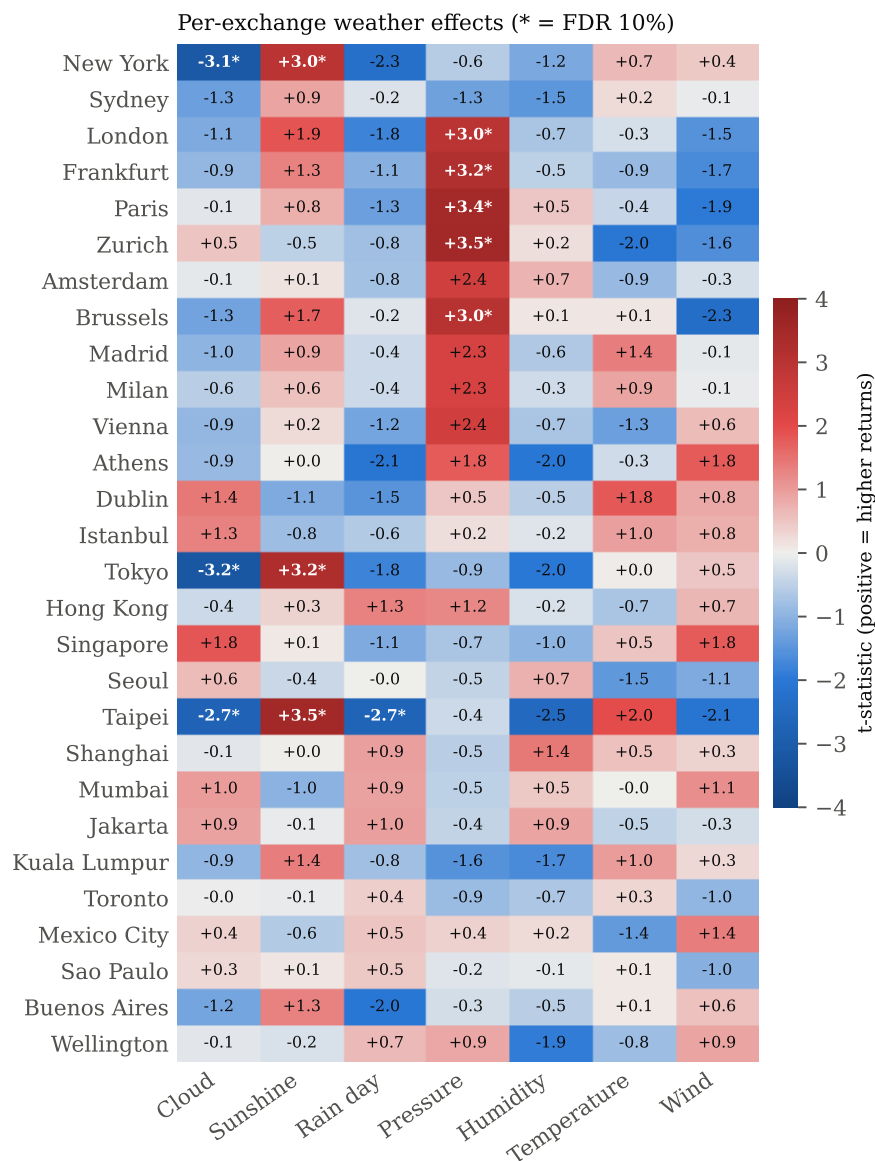


Figure 4: Exchange-level t -statistics for each weather variable (volatility-scaled returns, HAC standard errors). Positive values mean higher returns. Starred cells survive a Benjamini–Hochberg false-discovery rate of 10% across all 196 tests.

7 Identification and limitations

Weather is determined outside financial markets, so reverse causality is not a concern. But our regressions identify an *association* between local weather and returns, not the channel that produces it. At least three channels are consistent with the results: investor mood, the one the original literature proposed; physical effects on cash flows, which our industry results support for temperature (oil and utilities); and weather-driven changes in local activity or attention, such as fewer trades on rainy days. We do not separate these.

Several further limitations should be kept in mind. First, post-publication tests have limited power: the post-2003 US cloud interval $[-5.2, +1.3]$ bp is consistent with both persistence and disappearance. Second, ERA5 reanalysis and station records measure weather with error, which biases coefficients towards zero. Third, Study 2 uses price indices, and the Australian total return is approximated (Section 3). Fourth, the placebo tests have a resolution of 1/100 (Study 1) and 1/200 (Study 2), and cities that share weather systems, such as those in Europe, provide less independent evidence than their number suggests. Fifth, the Study 2 candidate list was compiled by the author, although it was fixed before any of its returns were examined. Finally, both pre-registrations were written and time-stamped by the author in version control. Their credibility rests on the public commit history.

8 Discussion and conclusion

Pre-registration and placebo inference change how the old weather results should be read. In our data, the New York cloud association is not an artefact of specification search. It replicates, survives multiplicity adjustment, and is more extreme than the same statistic computed with the weather of every one of 100 unrelated cities. But it is small, and most of the evidence for it comes from before 2007. After publication the US estimate is similar in size but imprecise, and the Australian estimate is close to zero, with a confidence interval tight enough to rule out effects of the size originally reported. The SAD cycle does not appear in Australia, and temperature matters only for industries whose cash flows depend on it. Study 2 generalises the picture. Across 26 further exchanges, fair weather (no rain, high pressure, sunshine) went with higher returns, and rain survives the full pre-registered decision rule. But the evidence for these effects comes mainly from before 2003, and the international cloud effect of [Hirshleifer and Shumway \(2003\)](#) is not detectable out of sample.

One possible interpretation is that the historical association reflected a localised mood channel, which mattered when trading was concentrated among people who experienced the local weather and has weakened as trading dispersed geographically and algorithmically. The decline in the Sydney coefficient, the near-zero New York estimates in windows after 2005, and the concentration of the Study 2 effects before 2003 fit this interpretation. However, our direct market-structure test (H5) is too imprecise to confirm it, and the evidence is equally consistent with the other channels discussed in Section 7. Future work could use investor-location data ([Goetzmann et al., 2015](#)) or intraday data around the electronic-trading transitions to sharpen the test.

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A Additional tables

Table 6: Robustness of the weather tests (exploratory; not multiplicity-adjusted).

Specification	Pred.	β (bp)	t	p	N
H1-US station (primary)	<0	-2.13	-2.72	0.003	17,736
H1-US ERA5 weather	<0	-1.94	-2.56	0.005	18,036
H1-US CHI weather (ERA5)	<0	-1.24	-1.61	0.054	18,036
H1-US full-sample deseasonalisation	<0	-2.07	-2.90	0.002	19,028
H1-US logit up/down	<0	-5.79	-3.73	<0.001	17,736
H1-US GARCH(1,1)-t mean	<0	-2.10	-4.24	<0.001	17,736
H1-AU station (primary)	<0	-1.77	-1.87	0.031	10,111
H1-AU ERA5 weather	<0	-1.47	-1.68	0.046	10,633
H1-AU MEL weather (ERA5)	<0	+0.24	+0.32	0.625	10,633
H1-AU full-sample deseasonalisation	<0	-2.03	-2.10	0.018	10,111
H1-AU AU price-only returns	<0	-1.76	-1.86	0.031	10,111
H1-AU logit up/down	<0	-4.11	-1.76	0.039	10,111
H1-AU GARCH(1,1)-t mean	<0	-1.05	-1.54	0.062	10,111
H2-US station (primary)	<0	-1.95	-1.18	0.119	17,736
H2-US ERA5 weather	<0	-1.40	-0.88	0.189	18,036
H2-US CHI weather (ERA5)	<0	-2.84	-1.69	0.046	18,036
H2-US full-sample deseasonalisation	<0	-1.97	-1.19	0.118	19,028
H2-US break at 2001-01 (HS working paper)	<0	-0.43	-0.27	0.395	17,736
H2-AU station (primary)	<0	+0.07	+0.06	0.523	10,111
H2-AU ERA5 weather	<0	-0.10	-0.10	0.462	10,633
H2-AU MEL weather (ERA5)	<0	-0.06	-0.06	0.477	10,633
H2-AU full-sample deseasonalisation	<0	-0.10	-0.08	0.469	10,111
H2-AU AU price-only returns	<0	+0.06	+0.05	0.521	10,111
H2-AU break at 2001-01 (HS working paper)	<0	-0.61	-0.54	0.295	10,111
H4-US station (primary)	<0	+0.13	+0.19	0.574	18,538
H4-US ERA5 weather	<0	+0.00	+0.00	0.502	18,036
H4-US CHI weather (ERA5)	<0	+0.74	+1.13	0.870	18,036
H4-US full-sample deseasonalisation	<0	-0.06	-0.09	0.463	19,938
H4-US logit up/down	<0	+2.09	+1.46	0.928	18,538
H4-US GARCH(1,1)-t mean	<0	+0.56	+1.28	0.899	18,538
H4-AU station (primary)	<0	+0.67	+0.85	0.803	10,610
H4-AU ERA5 weather	<0	+0.80	+0.98	0.837	10,633
H4-AU MEL weather (ERA5)	<0	+0.78	+1.02	0.847	10,633
H4-AU full-sample deseasonalisation	<0	+0.49	+0.60	0.725	10,610
H4-AU AU price-only returns	<0	+0.68	+0.87	0.808	10,610
H4-AU logit up/down	<0	+1.05	+0.52	0.700	10,610
H4-AU GARCH(1,1)-t mean	<0	+0.07	+0.12	0.547	10,610
RAIN-US (06-16h rain dummy)	<0	-5.42	-2.86	0.002	17,290
RAIN-AU (06-16h rain dummy)	<0	-0.00	-0.59	0.279	5,715

Table 7: Seasonal tests: SAD, FALL and the Halloween effect (exploratory).

Specification	Pred.	β (bp)	t	p	N
US SAD, 1955-2026	>0	+2.02	+2.40	0.008	18,036
US FALL, 1955-2026	<0	-2.55	-1.14	0.127	18,036
US SAD, 1928-2026	>0	+1.39	+1.85	0.032	25,866
US FALL, 1928-2026	<0	-3.01	-1.49	0.068	25,866
US SAD, post-2003	>0	+1.95	+1.09	0.139	5,829
US FALL, post-2003	<0	-3.76	-0.81	0.210	5,829
AU SAD (own season), alone	>0	+0.22	+0.20	0.419	10,633
AU SAD_NYC (calendar), alone	>0	-0.16	-0.19	0.576	10,633
AU Halloween (Nov-Apr), alone	>0	+1.38	+0.86	0.194	10,633
AU SAD with Halloween	>0	+1.55	+0.94	0.174	10,633
AU Halloween with SAD	>0	+2.87	+1.19	0.116	10,633
AU Halloween, pre-2003	>0	+1.61	+0.67	0.252	4,773
AU Halloween, post-2003	>0	+0.78	+0.38	0.353	5,860

Table 8: Market-structure and size tests (exploratory).

Specification	Pred.	β (bp)	t	p	N
US cloud x post-2020-03 floor closure	>0	-1.92	-0.58	0.718	17,736
US cloud, 1955-2006	<0	-2.05	-2.53	0.006	12,806
US cloud, 2007-2026	<0	-1.87	-0.97	0.166	4,930
AU small-minus-large ($\hat{A}XSO - \hat{A}FLI$) on cloud	<0	-1.40	-1.16	0.122	3,213

Table 9: US industry returns on weather anomalies, controlling for the same-day market return (exploratory; two-sided p).

Specification	Pred.	β (bp)	t	p	N
Industry Util (market-adjusted) on temp_z	$\pm$	-0.98	-2.25	0.025	18,538
Industry Util (market-adjusted) on cloud_z	$\pm$	+0.47	+1.02	0.309	17,736
Industry Oil (market-adjusted) on temp_z	$\pm$	-2.03	-2.87	0.004	18,538
Industry Oil (market-adjusted) on cloud_z	$\pm$	+0.76	+1.03	0.305	17,736
Industry Coal (market-adjusted) on temp_z	$\pm$	-0.17	-0.13	0.897	18,538
Industry Coal (market-adjusted) on cloud_z	$\pm$	+5.64	+3.82	<0.001	17,736
Industry Agric (market-adjusted) on temp_z	$\pm$	-0.94	-1.15	0.251	18,538
Industry Agric (market-adjusted) on cloud_z	$\pm$	+0.27	+0.30	0.768	17,736
Industry Food (market-adjusted) on temp_z	$\pm$	-0.51	-1.17	0.243	18,538
Industry Food (market-adjusted) on cloud_z	$\pm$	-0.22	-0.47	0.639	17,736
Industry Rtail (market-adjusted) on temp_z	$\pm$	+0.51	+1.19	0.233	18,538
Industry Rtail (market-adjusted) on cloud_z	$\pm$	-0.16	-0.35	0.725	17,736
Industry Meals (market-adjusted) on temp_z	$\pm$	+0.33	+0.56	0.574	18,538
Industry Meals (market-adjusted) on cloud_z	$\pm$	+0.06	+0.10	0.919	17,736
Industry Insur (market-adjusted) on temp_z	$\pm$	-0.77	-1.39	0.165	18,538
Industry Insur (market-adjusted) on cloud_z	$\pm$	-0.32	-0.54	0.589	17,736

Table 10: Sensitivity checks. Australian tests with price-only returns and with calendar-month fixed effects (which absorb any month-specific dividend add-back exactly), and standard-error bandwidths. Coefficients in bp (Study 2: % of a daily SD); one-sided p .

Check	β	t	p	N
corr(dividend add-back, cloud_z)	correlation -0.006			
corr(dividend add-back, temp_z)	correlation -0.011			
H1-AU total return (primary)	-1.77	-1.87	0.031	10,111
H1-AU price only	-1.76	-1.86	0.031	10,111
H1-AU total return + month FE	-1.85	-1.93	0.027	10,111
H2-AU total return (primary)	+0.07	+0.06	0.523	10,111
H2-AU price only	+0.06	+0.05	0.521	10,111
H2-AU total return + month FE	+0.01	+0.01	0.503	10,111
H3-AU total return (primary)	-1.38	-1.06	0.855	10,633
H3-AU price only	-0.36	-0.27	0.607	10,633
H4-AU total return (primary)	+0.67	+0.85	0.803	10,610
H4-AU price only	+0.68	+0.87	0.808	10,610
H4-AU total return + month FE	+0.64	+0.82	0.794	10,610
H1-US Newey-West lags=5	-2.13	-2.72	0.003	17,736
H1-US Newey-West lags=10	-2.13	-2.65	0.004	17,736
H1-US Newey-West lags=20	-2.13	-2.61	0.005	17,736
S6 Driscoll-Kraay lags=5	-1.55	-2.58	0.005	222,941
S6 Driscoll-Kraay lags=10	-1.55	-2.63	0.004	222,941
S6 Driscoll-Kraay lags=20	-1.55	-2.66	0.004	222,941

B Deviations from the pre-registration

Study 2 had no deviations from its pre-registration. For Study 1, six deviations were logged in `prereg/deviations.md`; the table there is authoritative. D1–D5 were made before the corresponding results were seen. D1: weather sources, because NOAA discontinued the ISD-Lite files in 2025. D2: at least three valid hourly reports per day instead of four, because Sydney reported three-hourly before 2000. D3: dividend add-back source, because the Reserve Bank of Australia table named in the plan no longer exists. D4: H3 judged on the Holm test alone, because no placebo can be defined for an astronomical variable. D5: ERA5 obtained from the Copernicus Climate Data Store rather than a rate-limited third-party API. D6 is a bug fix made after results were seen: the year-shifted placebo had dropped non-matching weekdays. It affects only that secondary statistic; both versions are archived.

C Study 2 exchange selection

The candidate list was compiled before any Study 2 return was examined. It contains the main national benchmark index of 37 exchanges in developed and large emerging markets for which we could identify a Yahoo Finance ticker. An exchange is eligible if:

1. its index has daily closing levels on Yahoo Finance starting on or before 2006-01-01;
2. the series continues through August 2026; and
3. fewer than 5% of its daily closes are unchanged from the previous close (a marker of stale prices).

New York and Sydney are carried over from Study 1 and excluded from the primary tests. The 26 eligible exchanges are listed in Section 6. The 11 excluded candidates, and the reasons, are: Stockholm (`OMX`, history from 2008), Copenhagen (`OMXC25`, 2016), Helsinki (`OMXH25`, 2013), Oslo (`OSEBX.OL`, 2013), Lisbon (`PSI20.LS`, 2013), Johannesburg (`J203.JO`, 2012), Tel Aviv (`TA125.TA`, 16% zero-change days), and Bangkok (`SET.BK`), Manila (`PSEI.PS`), Santiago (`IPSA`) and Cairo (`CASE30`), which have no usable daily history.

D Implementation details

Daily weather. Hourly station reports are first averaged within each clock hour, so that special reports do not receive extra weight. The daily value is then the mean over local clock hours [06:00, 16:00), with time zones and daylight saving from the IANA database. Station days with fewer than three valid hours are missing (deviation D2). ERA5 series are complete. For ERA5 accumulations (solar radiation, precipitation), the value stamped t covers the hour $[t - 1, t)$ and is assigned to that hour, so the daily window sums the hours ending 07:00–16:00.

Climatology and standardisation. For a daily variable X_d on day d with ISO week $w(d)$ (week 53 merged into week 52) and year $y(d)$, let $\mathcal{C}_d = \{d' : w(d') = w(d), y(d) - 30 \leq y(d') \leq y(d) - 1\}$. The anomaly is

$$z_d = \frac{X_d - \bar{X}_{\mathcal{C}_d}}{s_{\mathcal{C}_d}},$$

where $\bar{X}$ and s are the sample mean and standard deviation over $\mathcal{C}_d$. The anomaly is missing unless $\mathcal{C}_d$ contains at least five distinct years. It therefore uses no information from year $y(d)$ or later. The Study 2 rain variable is the indicator $\mathbf{1}\{P_d \geq 1 \text{ mm}\}$ minus its mean over $\mathcal{C}_d$ (not scaled). The replication in Table 1 instead uses a full-sample week-of-year mean and standard deviation, as in [Hirshleifer and Shumway \(2003\)](#).

Missing data. Regressions use listwise deletion: a day enters a regression only if the return, its two lags and every regressor are observed. Nothing is imputed.

Calendar controls. A trading day is pre-holiday (post-holiday) if at least one weekday with no trading follows (precedes) it before the next (previous) trading day. Holidays are therefore inferred from each index’s own trading calendar. Half-days and early closes cannot be identified in the data and are treated as ordinary trading days. The turn-of-year dummy covers the last trading day of December and the first five of January, and the tax-year-end dummy the last five trading days of December (US) or June (Australia).

Standard errors. Newey–West errors use a Bartlett kernel with five lags, one trading week. Daily index returns have little serial correlation, and Table 10 shows the H1-US t -statistic moves from -2.72 to -2.61 with 20 lags. Driscoll–Kraay errors sum the regression scores across exchanges on each date and apply the same Bartlett kernel with five lags to the resulting time series (S6 t -statistic -2.58 with 5 lags, -2.66 with 20).

Romano–Wolf. 5,000 replications of a joint stationary bootstrap with mean block length 10 trading days (seed 20260927), resampling the union of all test dates so that dependence across tests is preserved. For test j the bootstrap statistic is $\delta_j(\hat{\beta}_j^* - \hat{\beta}_j)/\hat{se}_j$, where δ_j is the predicted sign and $\hat{se}_j$ the original HAC standard error, followed by the usual stepdown.

Placebos. In Study 1 every one of the 100 pre-registered placebo cities is used once (a census, not a sample). The year-shifted placebo uses the full calendar-day weather series shifted by $k = 1, \dots, 30$ years (deviation D6). In Study 2 each of 200 draws (seed 20260928) assigns every exchange a city drawn uniformly, with replacement, from the placebo cities that host no Study 2 exchange and lie more than 1,000 km from that exchange. The same city may therefore be assigned to several exchanges in one draw.

Study 2 returns. Exchange-days with $|r| > 20\%$ or an exactly zero return are dropped. The scaling volatility $\sigma_{i,t-1}$ is the standard deviation of the previous 60 returns (at least 40), so it uses only information available before day t .

Software. Python 3.12 with NumPy, pandas, statsmodels and arch. All code, the pre-registrations and the deviation log are in the repository.